02. KnowBase AI

AI-Powered Personal Knowledge Management System

Concept: Users upload PDFs, documents, notes, and other files, then ask questions and get answers grounded in their uploaded content.

Features

- PDF and document uploads.
- AI-powered question answering.
- Semantic search.
- Source citations for answers.
- Personal knowledge collections.
- Document sharing.
- Conversation history.

Development stack: React, Node.js, Express, MongoDB, Cloudinary or object storage, embeddings API, vector database such as Qdrant.

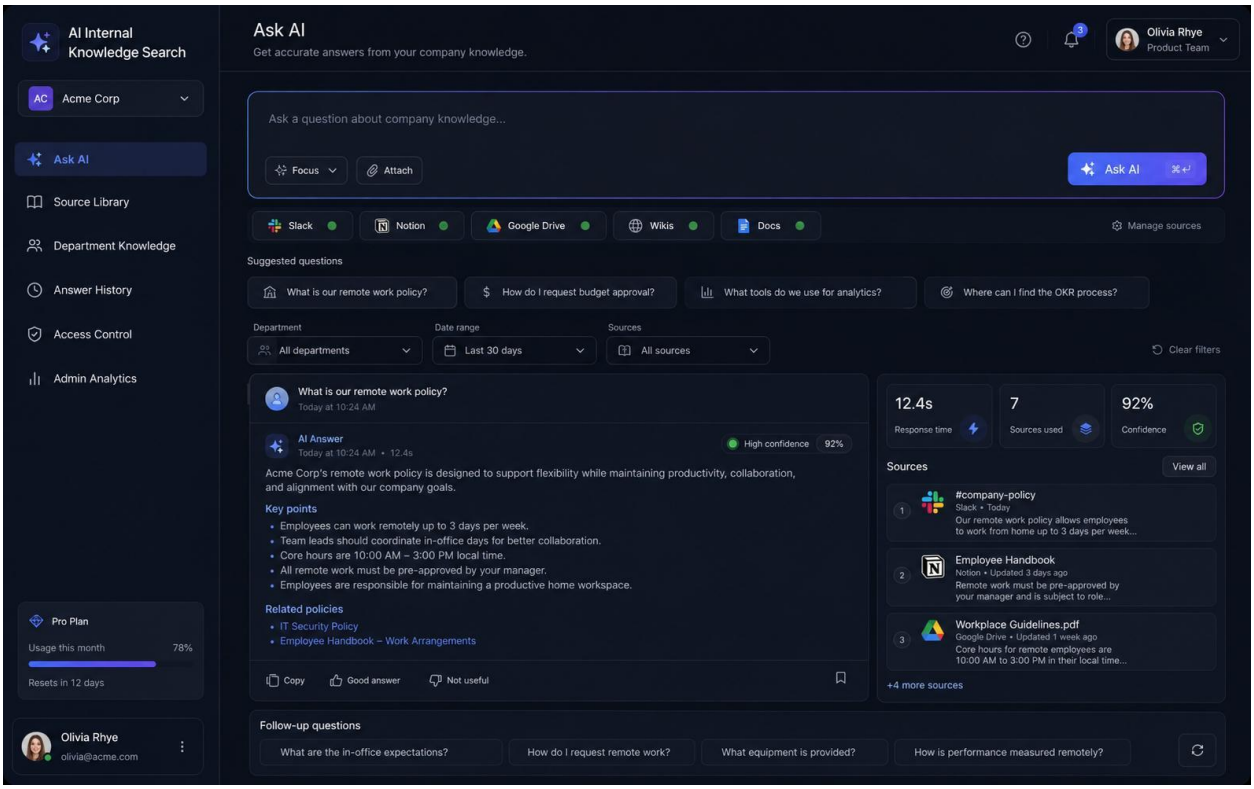
Database models: User, Document, Collection, Chunk, Conversation, Message.

Development phases

1. Build authentication and file upload.
2. Extract text from documents.
3. Split documents into chunks.
4. Generate embeddings and store vectors.
5. Implement retrieval-augmented generation (RAG).
6. Display answers with source references.

Placement value: AI integration, search algorithms, document processing, and database design.

Image:


AI Internal Knowledge Search
AC Acme Corp
Ask AI
Source Library
Department Knowledge
Answer History
Access Control
Admin Analytics
Pro Plan
Usage this month 78%
Resets in 12 days
Olivia Rhye
olivia@acme.com
Ask AI
Get accurate answers from your company knowledge.
3
Olivia Rhye
Product Team
Ask a question about company knowledge...
Focus
Attach
Ask AI ⌘↵
Slack
Notion
Google Drive
Wikis
Docs
Manage sources
Suggested questions
What is our remote work policy?
How do I request budget approval?
What tools do we use for analytics?
Where can I find the OKR process?
Department
All departments
Date range
Last 30 days
Sources
All sources
Clear filters
What is our remote work policy?
Today at 10:24 AM
AI Answer
Today at 10:24 AM • 12.4s
High confidence 92%
Acme Corp's remote work policy is designed to support flexibility while maintaining productivity, collaboration, and alignment with our company goals.
Key points
Employees can work remotely up to 3 days per week.
Team leads should coordinate in-office days for better collaboration.
Core hours are 10:00 AM – 3:00 PM local time.
All remote work must be pre-approved by your manager.
Employees are responsible for maintaining a productive home workspace.
Related policies
IT Security Policy
Employee Handbook – Work Arrangements
Copy
Good answer
Not useful
12.4s
Response time
7
Sources used
92%
Confidence
Sources
View all
1
#company-policy
Slack • Today
Our remote work policy allows employees to work from home up to 3 days per week...
2
Employee Handbook
Notion • Updated 3 days ago
Remote work must be pre-approved by your manager and is subject to role...
3
Workplace Guidelines.pdf
Google Drive • Updated 1 week ago
Core hours for remote employees are 10:00 AM to 3:00 PM in their local time...
+4 more sources
Follow-up questions
What are the in-office expectations?
How do I request remote work?
What equipment is provided?
How is performance measured remotely?